VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belgaum-590014

An Project Synopsis On

"FocusFlow AI – Smart Study & Distraction Detection System"

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By

KESHAV G K (1CE23CS070)
MARUTI J (1CE23CS088)
K S MANIKANTA (1CE23CS084)
RAJESH MURANAL (1CE23CS127)

Under the Guidance of
TARA V K & SHRUTI B S
Designation,Dept of CSE

CITY ENGINEERING COLLEGE

Doddakallasandra, Kanakapura Road,
Bengaluru-560061

Project Synopsis

Title of the Project
FocusFlow AI – Smart Study & Distraction Detection System

Abstract
This project aims to develop a smart productivity system that monitors user behavior during study sessions and detects distractions in real time. FocusFlow AI uses browser-based behavioral signals such as tab switching, typing activity, mouse movement, and idle time to analyze user focus levels. The system computes a dynamic Focus Score and provides real-time alerts when distractions are detected. Additionally, it offers personalized study suggestions and detailed productivity analytics through a web dashboard. By bridging the gap between time spent and actual focus, the system helps students and professionals improve their productivity and study habits while maintaining a privacy-first approach.

Problem Statement
Students and remote workers often get distracted by social media and multitasking, leading to reduced productivity. Existing tools mainly track time spent but fail to measure actual focus or detect distractions in real time. There is a need for an intelligent system that can monitor user behavior, identify distractions instantly, and provide meaningful insights to improve focus and efficiency.

Objectives

- To design a system that monitors user behavior during study sessions using browser-based tracking.
- To detect distractions in real time based on user activity patterns.
- To calculate a dynamic Focus Score representing the quality of concentration.
- To provide real-time alerts and notifications to improve user focus.
- To generate personalized study recommendations using behavioral data.
- To develop a web dashboard for visualizing productivity trends and analytics.

Methodology
The system is implemented as a browser-based application integrated with a web dashboard and backend server. A browser extension captures user activity such as tab switching, idle time, and input interactions. This data is sent to a backend server using WebSocket communication.
A Focus Engine processes the data using rule-based logic and lightweight machine learning techniques to classify user states as focused or distracted. A Focus Score is calculated using a weighted combination of behavioral signals. The system then triggers real-time alerts and sends insights to the dashboard.

The backend is developed using Node.js, while the frontend dashboard is built using React.js. A Python-based AI module can be integrated to enhance prediction and personalization. MongoDB is used for storing user session data, and Redis is used for real-time processing.

Requirements

- Hardware: Computer/Laptop with internet connection
- Software:
    - Frontend: Next.js, Tailwind CSS
    - Backend: Node.js, Express.js
    - Database: MongoDB
    - Browser Extension: JavaScript (Chrome Extension API)
    - AI Module: Python (Flask, Scikit-learn)
    - Tools: VS Code, GitHub, Postman

Expected Outcome

- The project will result in a fully functional web-based productivity system capable of:
- Detecting distractions in real time
- Calculating and displaying a Focus Score
- Providing instant alerts and feedback
- Generating daily and weekly productivity reports
- Offering personalized study suggestions
- The system is expected to improve user productivity, reduce distraction frequency, and help users develop better focus habits.